

Mini-Project 1

1 Modeling an Inverted Pendulum on a Cart

A pendulum of mass m and length ℓ is pinned to a cart of mass M that is free to translate along a horizontal track, as shown in Figure 1. Start the pendulum near the upright position, give it a small perturbation, and apply no force to the cart. The pendulum will fall and swing down, and the cart will move in response. Simulate long enough for the motion to develop, and plot the cart position $x(t)$ and the pendulum angle $\theta(t)$.

Build the model twice: once in Simulink, directly from the equations of motion, and once in Simscape Multibody, from physical components. The Simulink model carries through the remaining mini-projects. The Simscape model is here so that you have built one. In the report, the response plots belong in the body; put the Simulink and Simscape models in the appendix as large, clean figures.

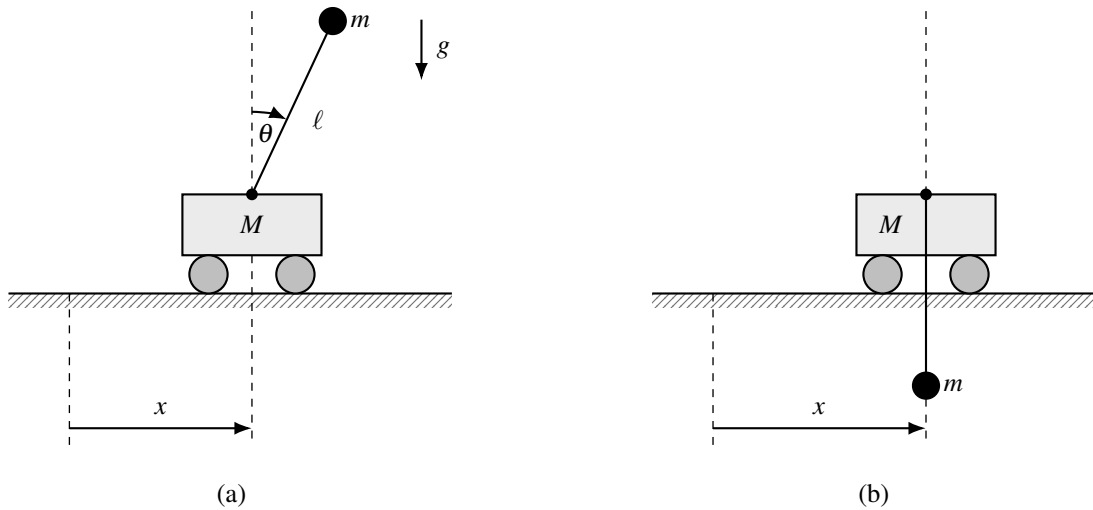


Figure 1: Inverted pendulum on a cart: (a) the starting condition near the upright position, and (b) a representative end condition, with the pendulum hanging below the pivot and the cart displaced. Contact between the pendulum and the ground is not modeled.

Table 1: Suggested parameters. Change them, or the assumptions behind them, if you have reason to.

Symbol	Description	Value
M	cart mass	2.0 kg
m	pendulum mass	0.5 kg
ℓ	pendulum length	1.0 m
c_θ	rotational damping at the pivot	0.01 N·m·s/rad
g	gravitational acceleration	9.81 m/s ²